



Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • goal: uncover spending patterns, segments, preferences, subscriptions

Dataset Snapshot

Rows

3,900

Columns

18 key features
(demographics, purchases,
behavior)

Missing Data

37 missing values in Review Rating

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3	14/100						BAG		
3	1P0						DIX		
4	116(00)						M		
14	708								
17	260						M		
15	110						M		
16	710						N		
16	430						N		
12	280						N		
16	220						M		
18	230								
15	150								
17	210								
13	460								
18	230						50		
15	110								
16	260								
10	160								
12	362								
14	250								
23	280								
21	990							O	
20									

Data Prep & Feature Engineering



01

Cleaning

Loaded with pandas; `df.info()` & `.describe()`

02

Imputation

Review Rating nulls filled with category median

03

Engineered

`age_group`, `purchase_frequency_days`

04

Standardized

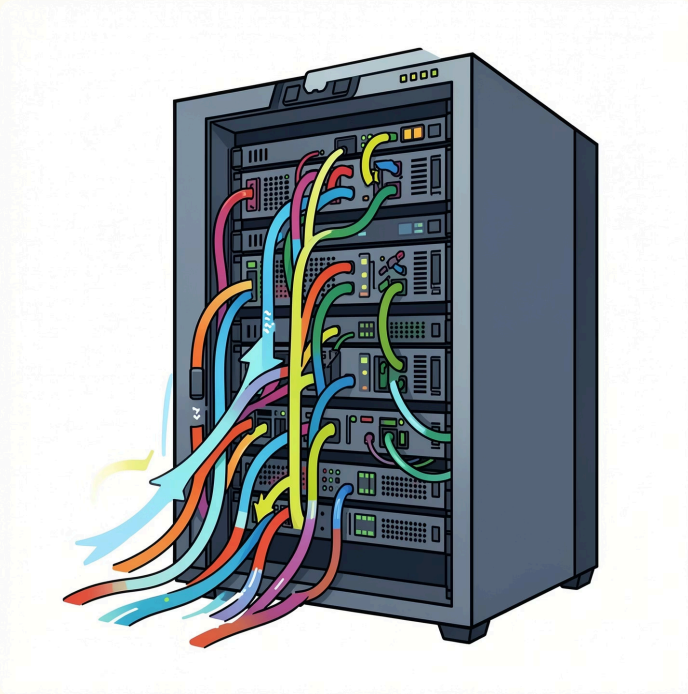
Columns renamed to `snake_case`

05

Pruned

Dropped `promo_code_used` (redundant with `discount_applied`)

Database Integration

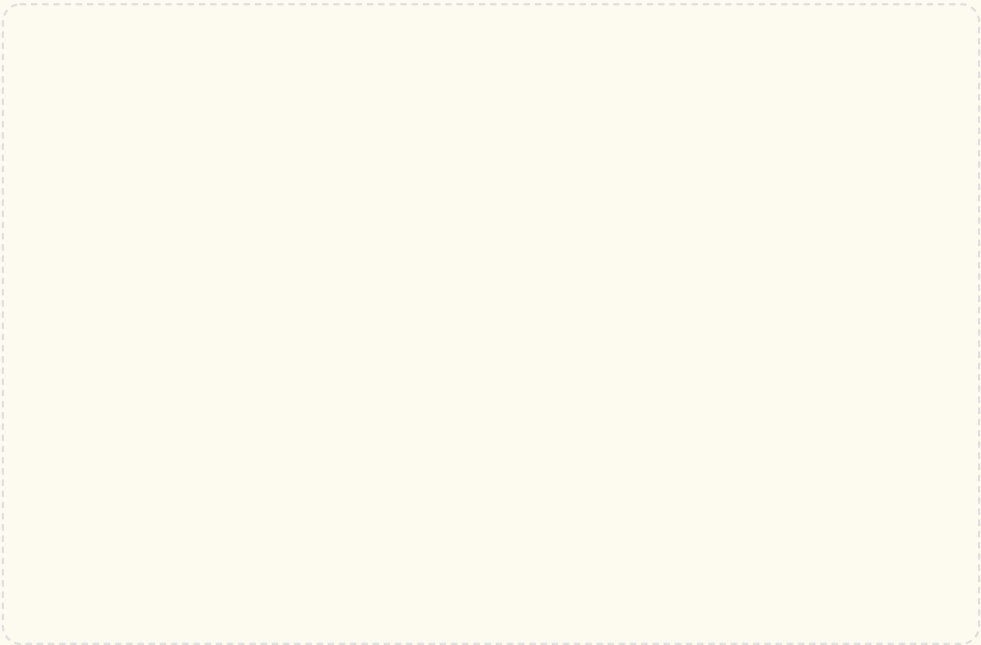


Cleaned DataFrame loaded to PostgreSQL

Enabled SQL analysis on business transactions

Revenue & Demographics (SQL)

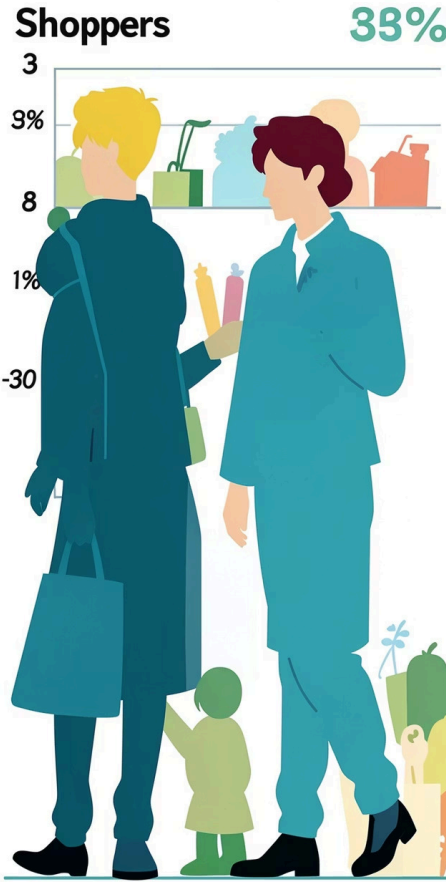
Compared total revenue by gender



Male

Gender

Female



Behavioral Insights



High-Spending Discount Users

Used discounts but spent above avg



Top Rated Products

Top 5 products by average rating



Shipping Comparison

Avg purchase: Standard vs Express

Segmentation & Loyalty



New

First-time purchasers



Returning

Multiple purchases over time



Loyal

>5 purchases — higher subscription likelihood

Product Performance



Top 3 per Category

Most purchased items listed by category



Discount-Dependent

5 products with highest % discounted purchases



Top Rated

Products with highest avg review ratings

Revenue by Age & Subscriptions

Age Group

18-24

25-34

35-44

45-54

55+

Compared total revenue by age group

Also compared subscribers vs non-subscribers: avg spend & total revenue



Dashboard & Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Review Discount Policy

Balance sales uplift with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express-shipping users